Strict bounds for packing 68 unit squares in a circle

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Abstract

Let $r(68)$ be the infimum of radii of disks containing 68 unit squares with arbitrary orientations and pairwise disjoint interiors. We give a rational-coordinate construction and a computer-assisted exclusion proving

$$\frac{39}{8} < r(68) < 5.031404629453352.$$

The construction has an exact enclosing radius specified by a quadratic equation with integer coefficients and an isolating interval of width 10^{-30}. The lower bound uses a nonnegative radial weight on 80 annuli. Its single-square inequality is verified on a complete subdivision of a continuous two-dimensional domain, using exact rational arithmetic and outward interval bounds. Coordinates, weights, subdivision data and verification programs accompany the paper. The exact optimum and feasibility at radius 5 remain open.

Keywords. Square packing; disk; exact construction; weighted area; computer-assisted proof.

1 Introduction and main result

Write $D_R = \{z \in \mathbb{R}^2 : \|z\| \le R\}$. A packing consists of closed unit squares whose interiors are pairwise disjoint; tangency is permitted. Translations and rotations of the squares are unrestricted. Translating the enclosing disk to the origin does not restrict the problem. We define $r(68)$ as the infimum of feasible radii.

Theorem 1. There is an explicitly specified packing of 68 unit squares with rational centers and rational unit direction vectors. If $\widehat{R}$ is its enclosing radius about the specified origin, then

$$\frac{39}{8} < r(68) \le \widehat{R} < U, \qquad U = \frac{5031404629453352}{10^{15}}. \tag{1}$$

Moreover,

$$\begin{aligned} &5.031404629453350730787899409097 < \widehat{R}, \\ &\widehat{R} < 5.031404629453350730787899409098. \end{aligned} \tag{2}$$

Montanher et al. [1] study the same freely rotating square-in-circle problem and give rigorous interval enclosures for the three-square optimum and its configurations. The distinction between approximate feasibility and proved feasibility is also central to Domes and Neumaier [2]. For a direct historical comparison at $n = 68$, Jelliss [3] describes a grid construction of radius $\sqrt{26}$. The upper bound in (1) is smaller, since $U^2 < 26$. This comparison does not assert a current record or priority over all published constructions.

The elementary area bound is $r(68) \ge \sqrt{68/\pi} \approx 4.65242649$; the present strict lower bound raises this baseline to $39/8 = 4.875$. Our contribution is an explicit exact construction together with a reproducible exclusion certificate over all real poses. Friedman's collection [4] provides further context for smaller instances; the version consulted lists $n \le 50$ and supplies no comparison at $n = 68$.

The lower-bound method assigns a minimum weighted area to every admissible square. Resource-counting arguments of this kind have predecessors in the unavoidable-set method of Kearney and Shiu [5]. A recent preprint of Karakuş [6] discusses more general nonnegative measures for packing bounds. Here the counting inequality itself is elementary; the substantive certificate verifies a particular radial weight for every real translation and orientation in a disk. Section 4 explains why its finite computations cover this continuous family.

2 An exact upper-bound certificate

2.1 Rational data and geometric verification

For each square, the data specify a center $c_i \in \mathbb{Q}^2$ and a vector $u_i \in \mathbb{Q}^2$ with $u_i \cdot u_i = 1$. Let $v_i = (-u_{i,2}, u_{i,1})$. Its four vertices are

$$c_i + \frac{\varepsilon u_i + \eta v_i}{2}, \qquad \varepsilon, \eta \in \{-1, 1\}. \tag{3}$$

Figure 1 shows the construction. It has a largely columnar interior and several rotated groups near the boundary. The exact data, rather than the plotted coordinates, define the squares.

For a pair $i < j$, put

$$H_{ij} = \frac{1 + |u_i \cdot u_j| + |\det(u_i, u_j)|}{2}.$$

On any axis $n \in \{u_i, v_i, u_j, v_j\}$, H_{ij} is the sum of the projected half-widths. Thus the squares have disjoint interiors whenever at least one such axis satisfies

$$|(c_i - c_j) \cdot n| \geq H_{ij}. \tag{4}$$

Conversely, the separating-axis theorem for two convex polygons shows that these four axes suffice. This can also be seen by examining the edge normals of their Minkowski difference. Equality in (4) allows boundary contact.

The accompanying verifier checks $u_i \cdot u_i = 1$ for all 68 squares, (4) for all $\binom{68}{2} = 2278$ pairs, and the containment of all 272 vertices. Each check is a rational comparison, hence reduces to integer arithmetic. Convexity of D_U then gives containment of every square. This proves the upper-bound assertion without relying on a geometric tolerance or on the numerical search that supplied the layout.

2.2 The exact radius of the displayed construction

For coprime positive integers p, q, the exact radius is

$$\widehat{R} = \sqrt{p/q}, \qquad \frac{p}{q} = \max_{i,\varepsilon,\eta} \left\| c_i + \frac{\varepsilon u_i + \eta v_i}{2} \right\|^2, \tag{5}$$

The maximum is attained only at square 25 with $(\varepsilon, \eta) = (1, 1)$. The complete integers, of 1451 and 1450 decimal digits respectively, are supplied in `upper/best-68-exact-radius.json`. Together with (2), the equation $qX^2 - p = 0$ specifies $\widehat{R}$ exactly. Squaring the rational endpoints verifies both strict inequalities in (2) and $\widehat{R} < U$.

The center in this calculation is fixed. We do not assert that it is a smallest enclosing circle of this point set when the center is allowed to move. In particular, $qX^2 - p$ is an equation for the displayed construction's radius, not a defining polynomial for the unknown optimum $r(68)$.

3 A weighted-area lower bound

Set

$$s = \frac{40}{39}, \qquad a = \frac{s}{2}, \qquad c = \frac{19926631}{20000000}. \tag{6}$$

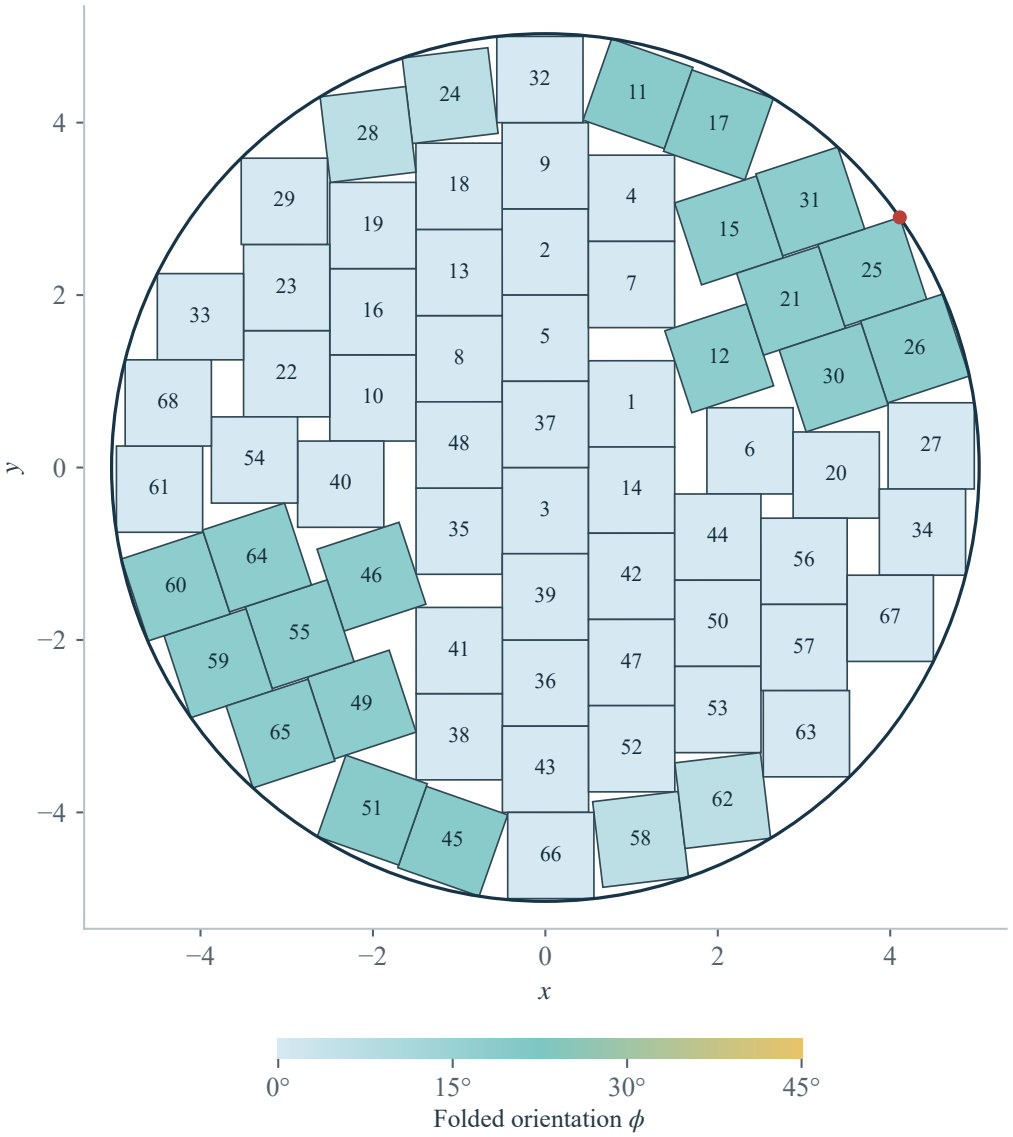


Figure 1. The certified 68-square construction. Labels refer to the exact coordinate file. Color records the folded orientation $\phi = \min(\alpha, 90^\circ - \alpha)$, where $\alpha \in [0, 90^\circ)$ is an edge angle modulo 90°. The red point is the unique farthest vertex, on square 25. All plotted coordinates are for display; the rational data define the packing.

We use a nonnegative radial function w, constant with value $w_j \in \mathbb{Q}_{\geq 0}$ on the annulus $j/16 \leq \|z\| < (j+1)/16$, for $j = 0, \ldots, 79$, and zero outside D_5. The 80 exact values are in `lower/weights.json`; Figure 2 displays them. Values on the bounding circles have no effect on the integrals.

Proposition 1. Every square S of side s contained in D_5, with arbitrary real center and orientation, satisfies

$$\int_S w(\|z\|)\, dz \geq c. \tag{7}$$

Proposition 1 is the computer-assisted step. Its certificate and the mathematical justification of its tests are given in Section 4.

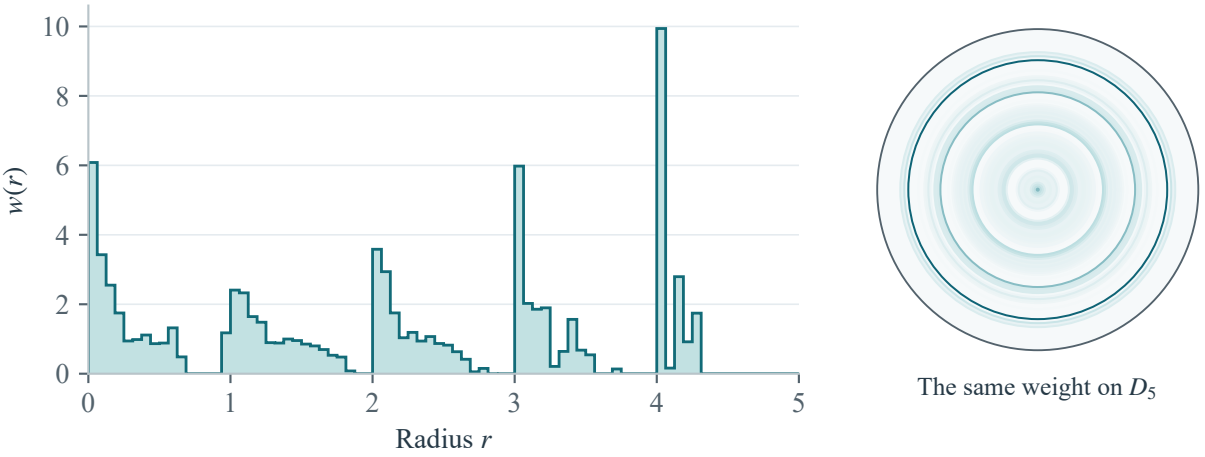


Figure 2. The 80-annulus radial weight, shown as a step function (left) and on the disk D_5 (right). Darker annuli carry larger weight. The exact breakpoints and heights are rational, and their oscillation is retained in the signed disk-area calculation.

The total weight is

$$\begin{aligned} C &= \int_{D_5} w(\|z\|)\,dz = \pi \sum_{j=0}^{79} w_j \frac{(j+1)^2 - j^2}{256} \\ &= \pi \frac{345050690237}{16000000000} < \frac{338752725}{5000000} < \frac{338752727}{5000000} = 68c. \end{aligned} \tag{8}$$

For completeness, the first strict inequality needs only

$$\pi < P := \frac{3141592653589794}{10^{15}}.$$

It follows from Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ and alternating-series bounds. Explicitly, if

$$T(q,m) = \sum_{k=0}^{m} \frac{(-1)^k}{(2k+1)q^{2k+1}},$$

then $\pi < 16T(5,12) - 4T(239,3) < P$; the latter comparison is rational. The gap between the last two fractions in (8) is $1/2500000$.

Suppose that 68 unit squares fit in a disk of radius at most $39/8$. Scaling the entire packing by $s = 40/39$ places 68 squares of side s inside D_5. Their interiors remain disjoint. Their boundaries have area zero, so nonnegativity and Proposition 1 imply

$$68c \le \sum_{i=1}^{68} \int_{S_i} w = \int_{\bigcup_i S_i} w \le C,$$

contradicting (8).

To obtain the strict inequality for the infimum, we also record attainment. Fix the feasible upper radius U. Every center in a packing inside D_U has norm at most $U - 1/2$, because its square contains a disk of radius $1/2$ about that center. Unit directions range over a compact circle. The vertex-containment conditions are closed, as is pairwise interior disjointness: the latter is a finite union of the closed conditions (4). The nonempty feasible set inside D_U is therefore compact. The continuous maximum of the vertex norms attains a minimum on it. Any packing that improves on U belongs to this set, so this minimum is $r(68)$. The exclusion just proved yields $r(68) > 39/8$ and completes Theorem 1 once Proposition 1 is verified.

4 Certifying all real placements

The certificate is a complete binary subdivision, rather than a set of sampled placements. Its leaves carry one of three tests. We give their soundness proofs and then describe the area kernel used to evaluate them.

4.1 Reduction and complete coverage

For a single square, rotate it about the origin until its sides are parallel to the coordinate axes. Both its containment in D_5 and its radial integral are invariant. Reflections and coordinate interchange then reduce its center to $x \geq y \geq 0$. This reduction is applied to the single-square integral, not separately to the squares in a packing. The admissible domain is exactly

$$\Omega = \{(x, y) : x \geq y \geq 0,\ (x + a)^2 + (y + a)^2 \leq 25\}. \tag{9}$$

The root box $B_0 = [0, 179/40]^2$ contains Ω, since $(179/40 + a)^2 + a^2 > 25$. Figure 3 shows the stored subdivision and a detail of its refinement. The image is a visualization of the certificate, not a replacement for its coverage and interval checks.

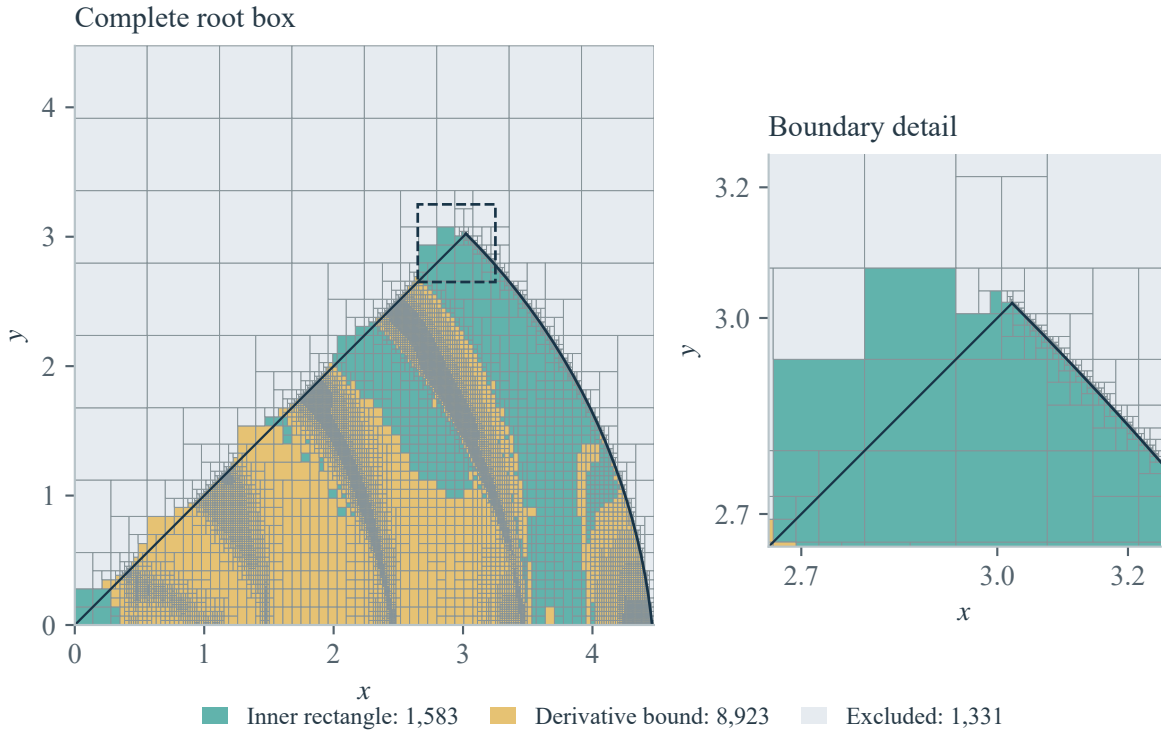


Figure 3. The complete binary certificate on B_0, colored by leaf test, with a magnified detail on the right. The dark line marks the circular and diagonal boundaries of Ω; gray boxes contain no admissible pose. The leaves are drawn directly from their stored binary paths. The exact paths and comparisons, rather than this visualization, determine coverage.

Bisect each box through the midpoint of its longer side, choosing the x direction in a tie. A binary word records a path from the root; both child boxes are closed. The supplied leaf words are distinct and prefix-free. If d is their maximum length, the verifier checks the exact identity

$$\sum_{\ell \text{ a leaf}} 2^{d-|\ell|} = 2^d. \tag{10}$$

Each word of length d then has exactly one leaf prefix. All depth-d boxes, and hence B_0, are covered, including shared boundaries.

Write a leaf box as $B = [b_0, b_1] \times [d_0, d_1]$. It contains no admissible point if

$$b_1 < d_0 \quad \text{or} \quad (b_0 + a)^2 + (d_0 + a)^2 > 25. \tag{11}$$

These strict rational tests do not discard a feasible boundary point.

4.2 A common inner rectangle

The intersection of all translated squares with centers in B is

$$K_B = [b_1 - a, b_0 + a] \times [d_1 - a, d_0 + a],$$

provided its two side lengths are positive. Since $w \geq 0$, every such square satisfies $\int_S w \geq \int_{K_B} w$. A rigorous lower endpoint for the latter integral that is at least c therefore certifies the whole box. This is the first acceptance test.

4.3 A derivative bound

Let $S(x, y) = [x - a, x + a] \times [y - a, y + a]$ and

$$A_r(x, y) = \operatorname{area}(S(x, y) \cap D_r).$$

With $h_r(e) = \sqrt{\max(0, r^2 - e^2)}$, define the chord-intersection length

$$J_r(e, z) = \max\{0, \min(z + a, h_r(e)) - \max(z - a, -h_r(e))\}. \tag{12}$$

This function is continuous, also at tangencies. Fubini's theorem and the fundamental theorem of calculus give

$$\begin{aligned} A_r(x, y) &= \int_{x-a}^{x+a} J_r(e, y)\, de, \\ \partial_x A_r(x, y) &= J_r(x + a, y) - J_r(x - a, y). \end{aligned} \tag{13}$$

Interchanging x, y gives the other partial derivative. Thus A_r is continuously differentiable on the whole plane.

For $z \geq 0$, $J_r(e, z)$ is nonincreasing in both $|e|$ and z: the former shrinks the centered disk chord, and the latter moves an interval of fixed length away from its center. Consequently, for $e \in [e_0, e_1]$ and $z \in [z_0, z_1] \subseteq [0, \infty)$,

$$J_r(e_{\text{far}}, z_1) \leq J_r(e, z) \leq J_r(e_{\text{near}}, z_0),$$

where $e_{\text{far}} = \max(|e_0|, |e_1|)$ and $e_{\text{near}} = \min_{e \in [e_0, e_1]} |e|$. Square-root enclosures and rational interval operations therefore give bounds for (13) throughout each box.

Put $w_{80} = 0$ and $r_j = j/16$. Almost everywhere,

$$w(\|z\|) = \sum_{j=1}^{80} (w_{j-1} - w_j) \mathbf{1}_{D_{r_j}}(z).$$

Hence

$$I(x, y) := \int_{S(x,y)} w = \sum_{j=1}^{80} (w_{j-1} - w_j) A_{r_j}(x, y) \tag{14}$$

is continuously differentiable. Some difference coefficients are negative; interval multiplication retains their signs. If $|\partial_x I| \leq M_x$ and $|\partial_y I| \leq M_y$ on B, with midpoint m and half-widths h_x, h_y, integration along the segment from m to any point of B gives

$$I(x, y) \geq \underline{I}(m) - M_x h_x - M_y h_y. \tag{15}$$

Here $\underline{I}(m)$ is a rigorous lower endpoint. If the right side is at least c, this second acceptance test certifies every real center in the box.

4.4 Certified disk–polygon intersection areas

It remains to enclose the integrals in the two acceptance tests. For $r > 0$ the continuous, locally Lipschitz vector field

$$V_r(z) = \begin{cases} z/2, & \|z\| \le r, \\ r^2 z/(2\|z\|^2), & \|z\| > r \end{cases}$$

has divergence $\mathbf{1}_{D_r}$ almost everywhere. For a counterclockwise polygon P, the divergence theorem therefore gives

$$\operatorname{area}(P \cap D_r) = \frac{1}{2} \sum_{[b,b+d] \subset \partial P} \det(b,d) \int_0^1 \min\left(1, \frac{r^2}{\|b+td\|^2}\right) dt. \tag{16}$$

The multiplier at the origin is interpreted by the continuous vector field. When $\det(b,d) = 0$, the edge contributes zero.

For other edges, $q(t) = \|b + td\|^2$ is a rational quadratic. Its extrema on a rational subinterval are determined exactly by its endpoints and, if present, its vertex. A segment wholly inside the disk contributes $\det(b,d)$ times half its parameter length. A segment wholly outside contributes $r^2/2$ times the signed angle between its endpoint vectors. Because the line does not pass through the origin, this angle lies strictly between $-\pi$ and π, and the usual atan2 branch is unambiguous.

To handle a crossing, outward square-root intervals enclose the roots of $q(t) = r^2$. Their rational endpoints, together with 0 and 1, subdivide the edge. A residual subinterval $[t_0, t_1]$ whose position relative to the circle remains uncertain is enclosed by the symmetric interval of radius

$$\frac{|\det(b,d)|}{2}(t_1 - t_0),$$

because the multiplier in (16) lies in $[0,1]$. Thus uncertain root locations are covered without an unproved inside/outside decision. Interval addition of the edge contributions encloses the area. Applying the signed combination in (14) encloses the weighted integral as well.

The implementation uses rational arithmetic for geometric decisions and Arb midpoint-radius intervals [7] at 256-bit working precision for square roots and angles. The proof depends on outward enclosure, not merely on the number of computed digits.

4.5 The finite certificate

The prefix test (10) verifies 11,837 leaves, with maximum depth 24. The complete replay gives the following partition:

Leaf test	Number
Common inner rectangle	1583
Derivative bound	8923
Excluded by (11)	1331
Total	11837

There are no pending leaves. Each acceptance test establishes (7) throughout its box; the excluded leaves contain no point of Ω. Complete coverage and rotational invariance prove Proposition 1 for all real poses.

5 Reproducibility and remaining questions

The accompanying supplement contains the complete rational coordinates, the exact integers p, q, the annular weights, all leaf paths, and the verification source. In the arXiv source archive these files are in the `anc/` directory; the file paths cited here are relative to that directory. The upper-bound verifier uses only Python's standard-library rational arithmetic. The lower-bound replay uses `python-flint 0.8.0` and the area kernel proved above. The revised verifier uses explicit acceptance checks, rejects optimized Python execution, and has no search-resume certification entry point. A fresh complete replay verifies all 11,837 leaves; its environment and source identities are recorded in the supplement. Separate instructions explain the required final success status; a partial run is not a certificate. Source and data SHA256 digests identify the supplied files. No tolerance-dependent feasibility test is used as an acceptance criterion.

These are computer-assisted proofs, with Python and FLINT/Arb in the trusted computational base; they are not proof-assistant formalizations. The strict lower bound applies to all real placements. The exact algebraic upper radius applies to one explicit construction. The interval still contains 5, so neither feasibility at that radius nor the exact value of $r(68)$ is decided.

AI assistance. Large language models assisted proof exploration, code preparation, literature review and drafting. The accompanying programs and mathematical arguments specify the evidence supporting the claims.

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